

Tweet Sentiment Classification

Apple & Google Products — NLP-Based Brand Monitoring

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Phase 4 Project | Moringa School | NLP Classification

Business Understanding

Problem Statement

Apple and Google receive thousands of tweets daily about their products. Manually reading and classifying them is impossible. Negative sentiment that goes undetected can damage brand reputation before the team has a chance to respond.

Solution

Build an NLP model that automatically classifies any tweet as Positive or Negative sentiment based on its text content alone — enabling real-time brand monitoring at scale.

Key Business Questions

1. What is the overall distribution of sentiment about Apple and Google products?
2. Which products attract the most positive vs. negative sentiment?
3. What words are most common in positive vs. negative tweets?
4. How does tweet length relate to emotional content?
5. Which model best classifies tweet sentiment on unseen data?

Data Understanding

9,093

Total Tweets

3

Columns

4

Sentiment Labels

63%

Product Column Missing

Data Source

Dataset: CrowdFlower Twitter Brand and Product Emotions

Source: data.world/crowdflower

Stakeholders

Stakeholder	Challenge They Face	How This Model Helps
Marketing Teams	Cannot track brand sentiment at scale	Monitor sentiment spikes around product launches in real time
Customer Experience	Negative tweets go unnoticed until viral	Detect and escalate negative tweets early
Product Development	Unclear which features customers dislike	Identify recurring pain points from negative tweets
PR & Communications	Slow response during reputation crises	Respond strategically with early sentiment signals
Social Media Analysts	Manual sentiment tagging is too slow	Automate tweet classification at any volume

Data Cleaning

1. Data Cleaning

- Removed 22 duplicate tweets
- Removed 1 missing text row
- Removed ambiguous labels (I can't tell)

3. TF-IDF Vectorization

- Unigrams only (ngram_range 1,1)
- Max 10,000 features
- Fit on training data only — no leakage

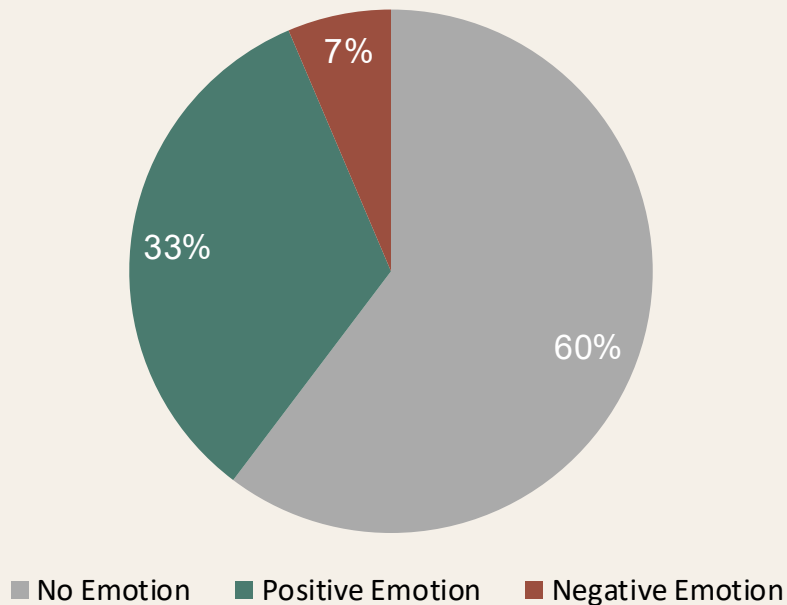
2. Text Preprocessing

- Lowercase + remove URLs, mentions, hashtags
- NLTK tokenization
- Stopword removal + Porter Stemming

4. Class Balancing

- Binary labels: Positive=1, Negative=0
- 5,956 samples after filtering
- SMOTE applied on training set only

Sentiment Distribution

**60.3%**

No Emotion

33.3%

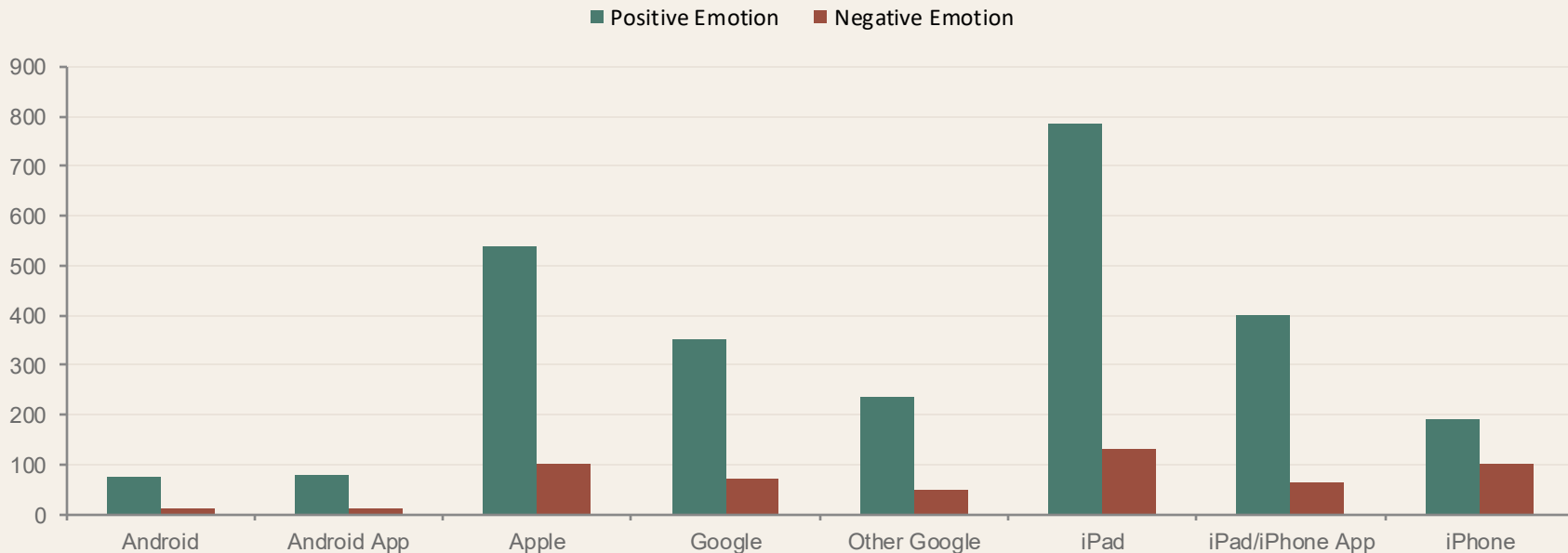
Positive

6.4%

Negative

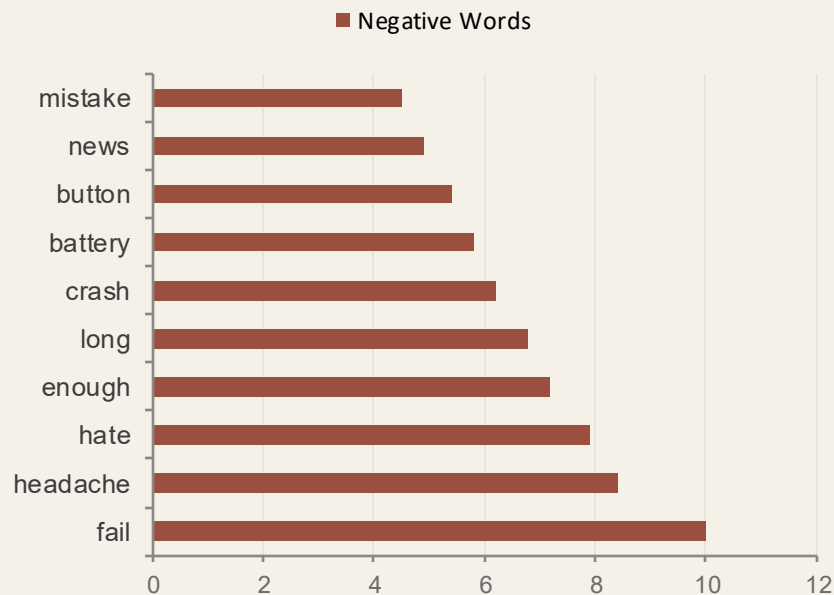
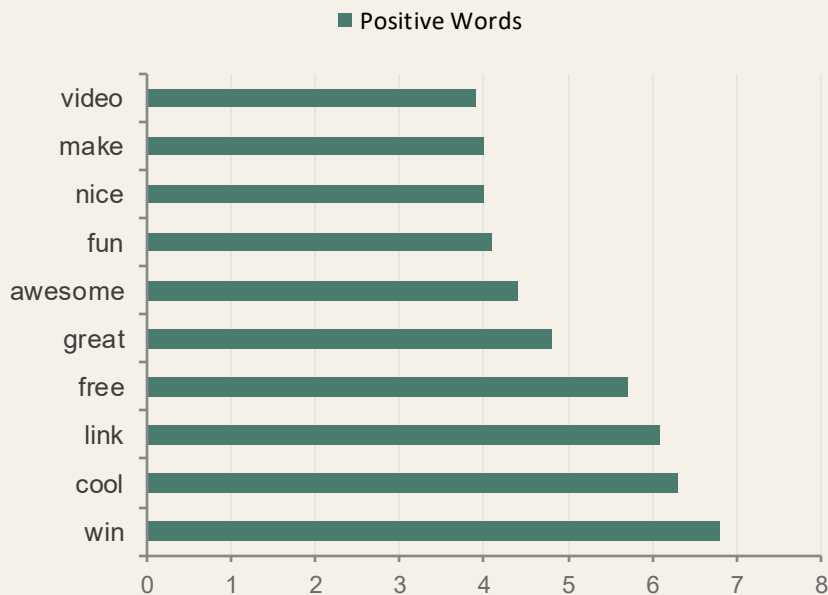
The dataset is heavily imbalanced. Only 6.4% of tweets are negative — the most business-critical class. SMOTE resampling was applied before modeling.

Sentiment by Product / Brand



iPad leads with ~785 positive and ~130 negative tweets. Apple brand follows. Android products barely register — the dataset skews heavily toward Apple users at SXSW.

Most Common Words by Sentiment Class



Vocabulary clearly separates the two classes. Positive tweets reflect excitement (win, free, great). Negative tweets point to specific frustrations (fail, crash, battery, hate).

Modeling

Model 1 — Multinomial Naive Bayes

Baseline

Fast probabilistic model, well-suited for TF-IDF count features. Assumes word independence. Used as baseline to beat.

Accuracy: 85% | F1 Macro: 0.76 | Neg. Recall: 69%

Model 2 — Logistic Regression

Fine-tuned with GridSearchCV

Linear model, does not assume word independence. Tuned: C, solver, max_iter using 3-fold cross-validation.

Accuracy: 86% | F1 Macro: 0.75 | Neg. Recall: 54%

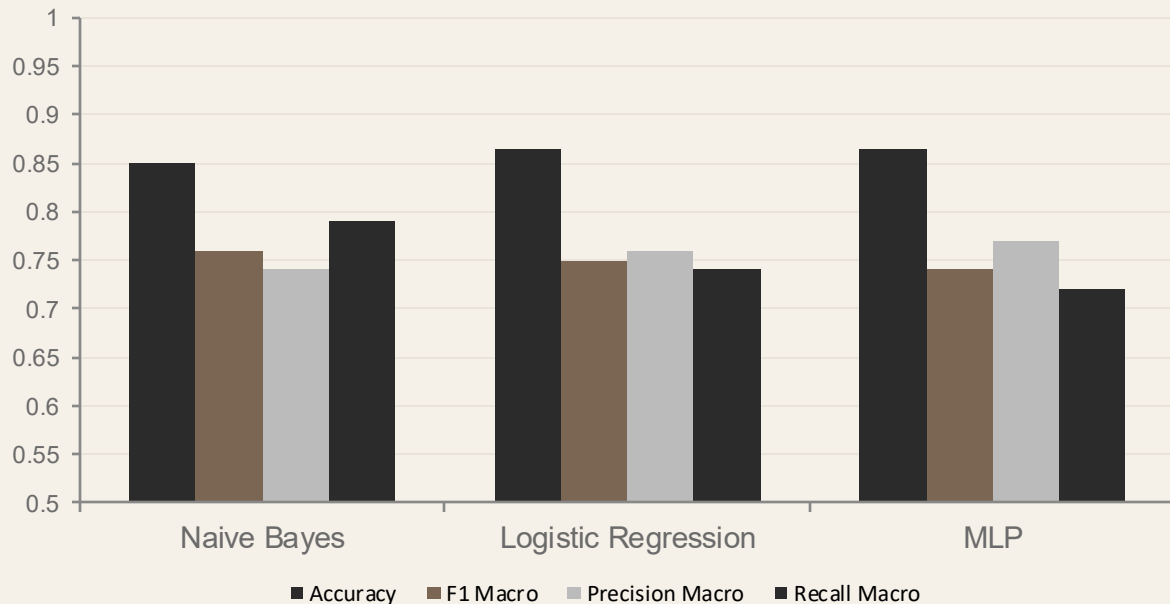
Model 3 — MLP Neural Network

Fine-tuned with GridSearchCV

Captures non-linear feature relationships. Requires dense TF-IDF array. Tuned: hidden layers, activation, alpha.

Accuracy: 86% | F1 Macro: 0.74 | Neg. Recall: 49%

Evaluation — Model Performance Comparison



Final Model

Naive Bayes

F1 Macro

0.76

Recall

0.79

Neg. Recall

69%

Naive Bayes achieves the highest macro F1 and recall — best at catching negative tweets, the most business-critical class.

Recommendations

01 Deploy Naive Bayes for Real-Time Monitoring

Connect the model and TF-IDF vectorizer to the Twitter/X API. Trigger alerts when negative sentiment exceeds a threshold.

02 Prioritize iPad and Apple Brand Monitoring

iPad and Apple brand generate the highest emotional volume. Set up hourly dashboards for these products specifically.

03 Implement Negative Keyword Alert System

Auto-escalate tweets containing fail, crash, battery, hate, or headache to customer experience teams immediately.

04 Expand to Multiclass Classification

60.3% of tweets are neutral and currently excluded. A multiclass model would unlock this data for richer brand insights.

05 Explore BERT-Based Models for Higher Accuracy

Transformer models pre-trained on Twitter data would better handle sarcasm and slang — recommended as the next phase.

Next Steps

- 1** → Deploy the Naive Bayes model via a FastAPI endpoint connected to the Twitter/X API for real-time sentiment monitoring.
- 2** → Expand the classifier to multiclass — include neutral tweets (currently 60.3% of data) for richer brand insights.
- 3** → Fine-tune a BERT-based transformer model on Twitter data to improve handling of sarcasm, slang, and class imbalance.

Thank You

Questions welcome

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